*5.4 The rectangular loop shown in Fig. P5.4 consists of 20 closely wrapped turns and is hinged along the z axis. The plane of the loop makes an angle of 30° with the y axis, and the current in the windings is 0.5 A. What is the magnitude of the torque exerted on the loop in the presence of a uniform field $\mathbf{B} = \hat{\mathbf{y}}2.4$ T? When viewed from above, is the expected direction of rotation clockwise or counterclockwise?

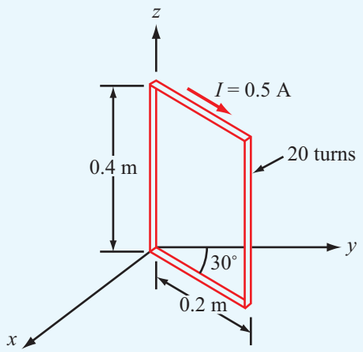


Figure P5.4 Hinged rectangular loop of Problem 5.4.

5.5 In a cylindrical coordinate system, a 2-m long straight wire carrying a current of 5 A in the positive z direction is located at $r = 4$ cm, $\phi = \pi/2$, and $-1 \text{ m} \le z \le 1$ m.

*(a) If $\mathbf{B} = \hat{\mathbf{r}}0.2\cos\phi$ (T), what is the magnetic force acting on the wire?

(b) How much work is required to rotate the wire once about the z axis in the negative ϕ direction (while maintaining $r = 4$ cm)?

(c) At what angle ϕ is the force a maximum?

5.6 A 20-turn rectangular coil with sides $l = 30$ cm and $w = 10$ cm is placed in the y–z plane as shown in Fig. P5.6.

(a) If the coil, which carries a current $I = 10$ A, is in the presence of a magnetic flux density

$$\mathbf{B} = 2 \times 10^{-2}(\hat{\mathbf{x}} + \hat{\mathbf{y}}2) \quad \text{(T)}.$$

Determine the torque acting on the coil.

(b) At what angle ϕ is the torque zero?

(c) At what angle ϕ is the torque maximum? Determine its value.

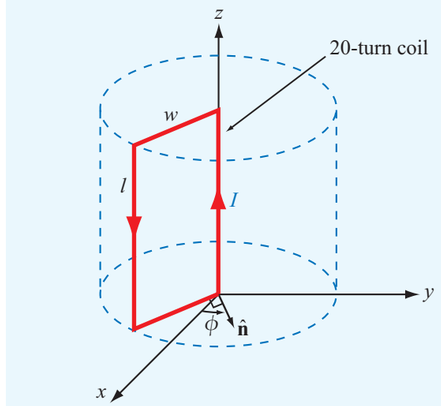


Figure P5.6 Rectangular loop of Problem 5.6.

Section 5-2: The Biot–Savart Law

*5.7 An 8 cm × 12 cm rectangular loop of wire is situated in the x–y plane with the center of the loop at the origin and its long sides parallel to the x axis. The loop has a current of 50 A flowing clockwise (when viewed from above). Determine the magnetic flux density at the center of the loop.

5.8 Use the approach outlined in Example 5-2 to develop an expression for the magnetic field $\mathbf{H}$ at an arbitrary point P due to the linear conductor defined by the geometry shown in Fig. P5.8. If the conductor extends between $z_1 = 3$ m and $z_2 = 7$ m and carries a current $I = 15$ A, find $\mathbf{H}$ at $P = (2, \phi, 0)$.

*5.9 The loop shown in Fig. P5.9 consists of radial lines and segments of circles whose centers are at point P. Determine the magnetic field $\mathbf{H}$ at P in terms of a, b, θ, and I.

5.10 An infinitely long, thin conducting sheet defined over the space $0 \le x \le w$ and $-\infty \le y \le \infty$ is carrying a current with a uniform surface current density $\mathbf{J}_s = \hat{\mathbf{y}}5$ (A/m). Obtain an expression for the magnetic field at point $P = (0, 0, z)$ in Cartesian coordinates.

*5.11 An infinitely long wire carrying a 25 A current in the positive x direction is placed along the x axis in the vicinity of a 20-turn circular loop located in the x–y plane (Fig. P5.11). If the magnetic field at the center of the loop is zero, what is the direction and magnitude of the current flowing in the loop?

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*5.4 图 P5.4 所示的矩形环由 20 个紧密缠绕的匝组成，并沿 z 轴铰接。环路平面与 y 轴成 30° 角，绕组中的电流为 0.5 A。在存在均匀磁场 B = ŷ 2.4 T 的情况下，施加在环路上的扭矩大小是多少？从上方观察时，预期的旋转方向是顺时针还是逆时针？

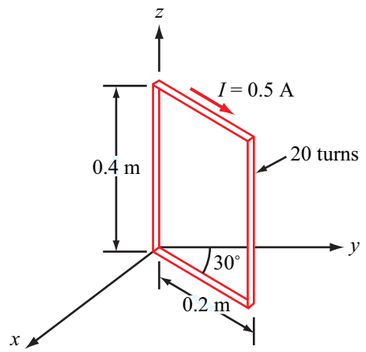


图 P5.4 问题 5.4 的铰接矩形环。

5.5 在圆柱坐标系中，在 z 正方向上承载 5 A 电流的 2 米长直导线位于 r = 4 cm、ϕ = π/2 和 −1 m ≤ z ≤ 1 m 处。

*(a) 如果 B = r̂ 0.2 cosϕ (T)，则作用在导线上的磁力是多少？ (b) 将导线绕 z 轴沿负 ϕ 方向旋转一次需要多少功（同时保持 r = 4 cm）？

(c) 在什么角度 ϕ 时力最大？

5.6 将边长 l = 30 cm、w = 10 cm 的 20 匝矩形线圈放置在 y-z 平面上，如图 P5.6 所示。

(a) 如果承载电流 I = 10 A 的线圈存在磁通密度

$$\mathbf{B} = 2 \times 10^{-2}(\hat{\mathbf{x}} + \hat{\mathbf{y}}2) \quad \text{(T)}.$$

确定作用在线圈上的扭矩。

(b) 在什么角度ϕ处扭矩为零？

(c) 在什么角度ϕ时扭矩最大？确定其价值。

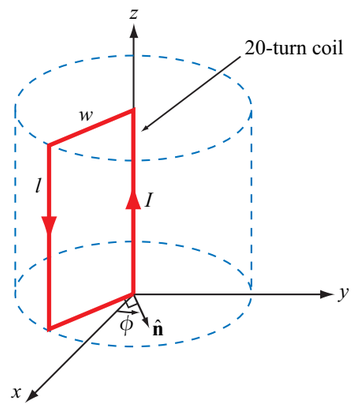


图 P5.6 问题 5.6 的矩形环。

第 5-2 节：毕奥-萨伐尔定律

*5.7 将一个 8 厘米 × 12 厘米的矩形线圈放置在 x-y 平面上，线圈的中心位于原点，其长边平行于 x 轴。该环路有 50 A 的电流顺时针流动（从上方观察）。确定环路中心的磁通密度。

5.8 使用例 5-2 中概述的方法，建立由图 P5.8 所示的几何形状定义的线性导体在任意点 P 处的磁场 H 的表达式。如果导体在 $z_1 = 3$ m 和 $z_2 = 7$ m 之间延伸并承载电流 I = 15 A，则在 P = (2、ϕ、0) 处找到 H。

*5.9 图 P5.9 所示的环路由径向线和圆心位于 P 点的圆段组成。用 a、b、θ 和 I 确定 P 处的磁场 H。

5.10 定义在空间 0 ≤ x ≤ w 和 −∞ ≤ y ≤ ∞ 上的无限长、薄的导电片承载着均匀表面电流密度 $J_s = \hat{y}5$ (A/m) 的电流。获得笛卡尔坐标中 P = (0, 0, z) 点的磁场表达式。

*5.11 将一根在 x 正方向上承载 25 A 电流的无限长导线沿 x 轴放置在位于 x-y 平面的 20 匝圆环附近（图 P5.11）。如果环路中心的磁场为零，则环路中流动的电流的方向和大小是多少？

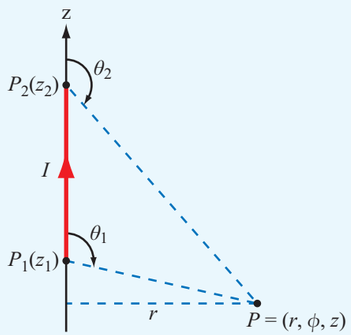


Figure P5.8 Current-carrying linear conductor of Problem 5.8.

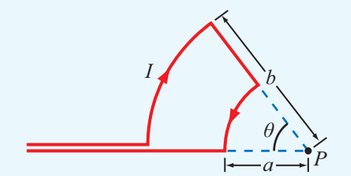


Figure P5.9 Configuration of Problem 5.9.

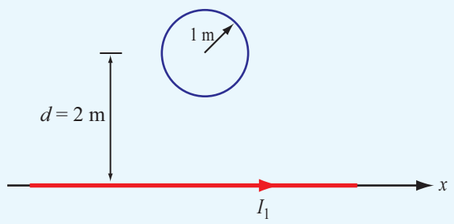


Figure P5.11 Circular loop next to a linear current (Problem 5.11).

5.12 Two infinitely long, parallel wires are carrying 6 A currents in opposite directions. Determine the magnetic flux density at point P in **Fig. P5.12**.

*5.13 A long, East–West-oriented power cable carrying an unknown current I is at a height of 8 m above the Earth's surface. If the magnetic flux density recorded by a magnetic-field meter placed at the surface is 15 μT when the current is flowing through the cable and 20 μT when the current is zero, what is the magnitude of I?

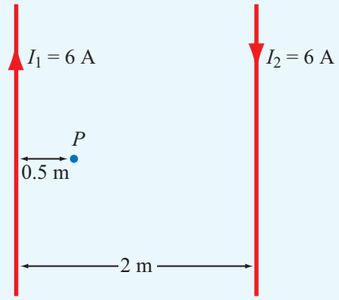


Figure P5.12 Arrangement for Problem 5.12.

5.14 Two parallel, circular loops carrying a current of 40 A each are arranged as shown in **Fig. P5.14**. The first loop is situated in the x–y plane with its center at the origin, and the second loop's center is at $z = 2$ m. If the two loops have the same radius $a = 3$ m, determine the magnetic field at

(a) $z = 0$

(b) $z = 1$ m

(c) $z = 2$ m

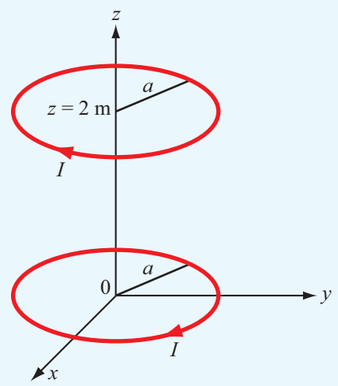


Figure P5.14 Parallel circular loops of Problem 5.14.

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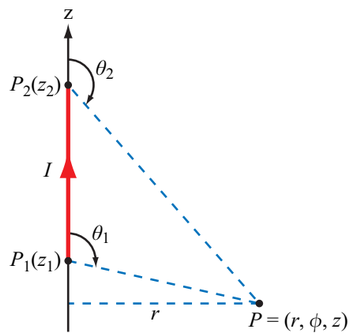


图 P5.8 问题 5.8 的载流线性导体。

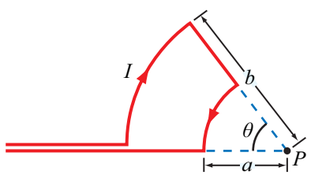


图 P5.9 问题 5.9 的配置。

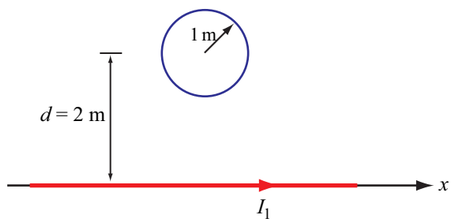


图 P5.11 线性电流旁边的圆形环路（问题 5.11）。

5.12 两根无限长的平行电线在相反方向上承载 6 A 电流。确定图 P5.12 中 P 点的磁通密度。

*5.13 一条东西向的长电力电缆，载有未知电流 I，位于距地球 8 m 的高度表面。如果当电流流过电缆时，放置在表面的磁场计记录的磁通密度为 15 μT，当电流为零时为 20 μT，那么 I 的大小是多少？

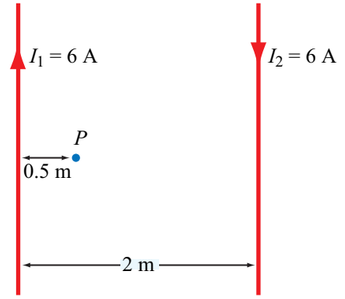


图 P5.12 问题 5.12 的排列。

5.14 两个并联的圆形回路，每个回路承载 40 A 的电流，如图 P5.14 所示布置。第一个环路位于 x-y 平面内，其中心位于原点，第二个环路的中心位于 z = 2 m。如果两个环路具有相同的半径 a = 3 m，则确定磁场

(a) z = 0 (b
) z = 1 m (c
) z = 2 m

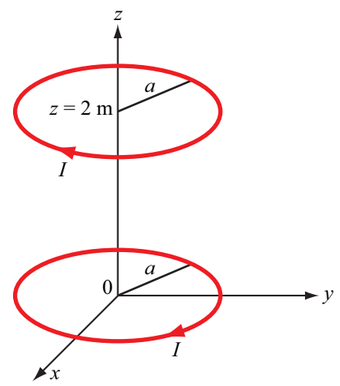


图 P5.14 问题 5.14 的平行循环。

5.15 A circular loop of radius a carrying current I_1 is located in the x–y plane as shown in **Fig. P5.15**. In addition, an infinitely long wire carrying current I_2 in a direction parallel with the z axis is located at $y = y_0$.

(a) Determine **H** at $P = (0, 0, h)$.

(b) Evaluate **H** for $a = 3$ cm, $y_0 = 10$ cm, $h = 4$ cm, $I_1 = 10$ A, and $I_2 = 20$ A.

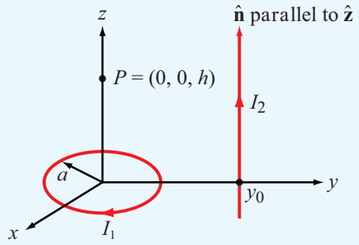


Figure P5.15 Problem 5.15.

*__5.16__ The long, straight conductor shown in **Fig. P5.16** lies in the plane of the rectangular loop at a distance $d = 0.1$ m. The loop has dimensions $a = 0.2$ m and $b = 0.5$ m, and the currents are $I_1 = 20$ A and $I_2 = 30$ A. Determine the net magnetic force acting on the loop.

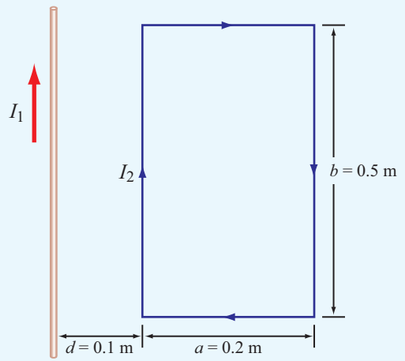


Figure P5.16 Current loop next to a conducting wire (Problem 5.16).

5.17 In the arrangement shown in **Fig. P5.17**, each of the two long, parallel conductors carries a current I, is supported by 8-cm long strings, and has a mass per unit length of 1.2 g/cm. Due to the repulsive force acting on the conductors, the angle θ between the supporting strings is 10°. Determine the magnitude of I and the relative directions of the currents in the two conductors.

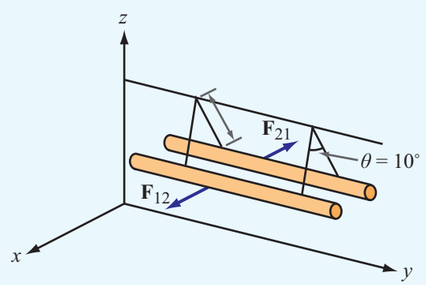


Figure P5.17 Parallel conductors supported by strings (Problem 5.17).

5.18 An infinitely long, thin conducting sheet of width w along the x direction lies in the x–y plane and carries a current I in the $-y$ direction. Determine the following:

*__(a)__ The magnetic field at a point P midway between the edges of the sheet and at a height h above it (**Fig. P5.18**).

(b) The force per unit length exerted on an infinitely long wire passing through point P and parallel to the sheet if the current through the wire is equal in magnitude but opposite in direction to that carried by the sheet.

5.19 Three long, parallel wires are arranged as shown in **Fig. P5.19**. Determine the force per unit length acting on the wire carrying I_3.

*__5.20__ A square loop placed as shown in **Fig. P5.20** has 2 m sides and carries a current $I_1 = 5$ A. If a straight, long conductor carrying a current $I_2 = 10$ A is introduced and placed just above the midpoints of two of the loop's sides, determine the net force acting on the loop.

5.15 载有电流 I_1 的半径为 a 的圆环位于 x-y 平面内，如图 P5.15 所示。此外，在与 z 轴平行的方向上承载电流 I_2 的无限长导线位于 $y = y_0$ 处。

(a) 确定 P = (0, 0, h) 处的 H。 (b) 评估 = 3 cm、$y_0 = 10$ cm、h = 4 cm、$I_1 = 10$ A 和 $I_2 = 20$ A 的 H。

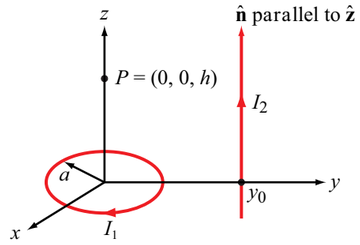


图 P5.15 问题 5.15。

*5.16 图 P5.16 所示的长直导体位于矩形环平面内，距离 d = 0.1 m。该环路的尺寸为 a = 0.2 m 和 b = 0.5 m，电流为 $I_1 = 20$ A 和 $I_2 = 30$ A。确定作用在环路上的净磁力。

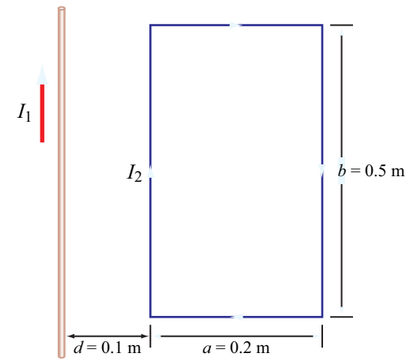


图 P5.16 导线旁边的电流环路（问题 5.16）。

5.17 在图 P5.17 所示的布置中，两个长的平行导体中的每一个都承载电流 I，由 8 厘米长的细绳支撑，单位长度的质量为 1.2 克/厘米。由于排斥力作用在导体上，支撑弦之间的角度θ为10°。确定 I 的大小以及两个导体中电流的相对方向。

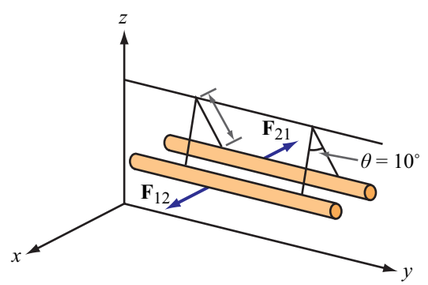


图 P5.17 由串支撑的平行导体（问题 5.17）。

5.18 沿 x 方向宽度为 w 的无限长、薄导电片位于 x-y 平面内，并在 −y 方向上承载电流 I。确定以下内容：

*(a) 片材边缘中间 P 点及其上方高度 h 处的磁场（图 P5.18）。

(b) 如果通过导线的电流大小相等但方向与片材所承载的电流方向相反，则施加在穿过点 P 并平行于片材的无限长导线上的每单位长度的力。

5.19 三根长的平行导线的排列如图 P5.19 所示。确定作用在承载 I_3 的导线上的每单位长度的力。

*5.20 如图 P5.20 所示放置的方形环路有 2 m 边，承载电流 $I_1 = 5$ A。如果引入承载电流 $I_2 = 10$ A 的直长导体并将其放置在环路两侧中点的正上方，请确定作用在环路上的合力。

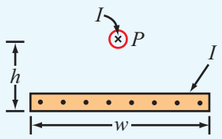


Figure P5.18 A linear current source above a current sheet (Problem 5.18).

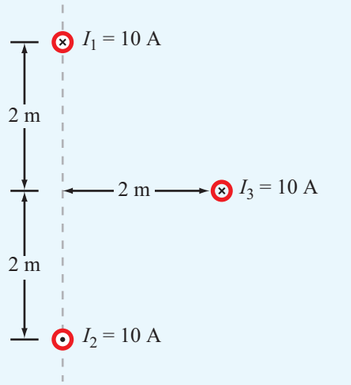


Figure P5.19 Three parallel wires of Problem 5.19.

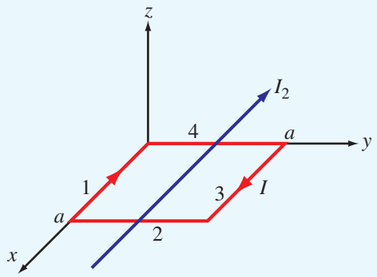


Figure P5.20 Long wire carrying current I_2, just above a square loop carrying I_1 (Problem 5.20).

Section 5-3: Maxwell's Magnetostatic Equations

5.21 Current I flows along the positive z direction in the inner conductor of a long coaxial cable and returns through the outer conductor. The inner conductor has radius a, and the inner and outer radii of the outer conductor are b and c, respectively.

(a) Determine the magnetic field in each of the following regions: $0 \le r \le a$, $a \le r \le b$, $b \le r \le c$, and $r \ge c$.

(b) Plot the magnitude of $\mathbf{H}$ as a function of r over the range from $r = 0$ to $r = 10$ cm given that $I = 10$ A, $a = 2$ cm, $b = 4$ cm, and $c = 5$ cm.

5.22 A long cylindrical conductor whose axis is coincident with the z axis has a radius a and carries a current characterized by a current density $\mathbf{J} = \hat{\mathbf{z}} J_0/r$, where J_0 is a constant and r is the radial distance from the cylinder's axis. Obtain an expression for the magnetic field $\mathbf{H}$ for

(a) $0 \le r \le a$

(b) $r > a$

5.23 Repeat Problem 5.22 for a current density $\mathbf{J} = \hat{\mathbf{z}} J_0 e^{-r}$.

*5.24 In a certain conducting region, the magnetic field is given in cylindrical coordinates by

$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{4}{r}[1 - (1 + 3r)e^{-3r}].$$

Find the current density $\mathbf{J}$.

5.25 A cylindrical conductor whose axis is coincident with the z axis has an internal magnetic field given by

$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{2}{r}[1 - (4r + 1)e^{-4r}] \quad \text{(A/m)} \quad \text{for } r \le a,$$

where a is the conductor's radius. If $a = 5$ cm, what is the total current flowing in the conductor?

Section 5-4: Vector Magnetic Potential

5.26 With reference to **Fig. 5-10**:

*(a) Derive an expression for the vector magnetic potential $\mathbf{A}$ at a point P located at a distance r from the wire in the x–y plane.

(b) Derive $\mathbf{B}$ from $\mathbf{A}$. Show that your result is identical with the expression given by Eq. (5.29), which was derived by applying the Biot–Savart law.

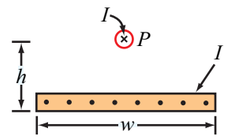


图 P5.18 电流片上方的线性电流源（问题 5.18）。

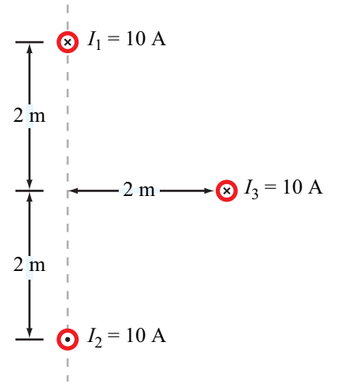


图 P5.19 问题 5.19 的三根平行线。

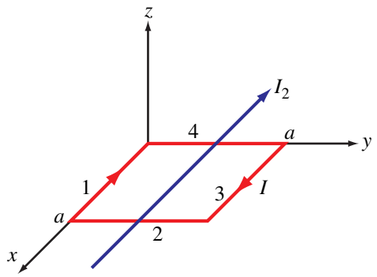


图 P5.20 承载电流 I_2 的长导线，位于承载 I_1（ 的方形环上方 问题 5.20)。

第 5-3 节：麦克斯韦静磁方程

5.21 电流 I 在长同轴电缆的内导体中沿 z 轴正方向流动，并通过外导体返回。内导体的半径为a，外导体的内半径和外半径分别为b和c。

(a) 确定以下每个区域的磁场：$0 \le r \le a$、$a \le r \le b$、$b \le r \le c$ 和 $r \ge c$。 (b) 假定 $I = 10$ A、$a = 2$ cm、$b = 4$ cm 和 $c = 5$ cm，在 $r = 0$ 到 $r = 10$ cm 范围内绘制 H 的幅度作为 r 的函数。

5.22 轴线与 z 轴重合的长圆柱形导体，其半径为 a，并承载由电流密度 $J = \hat{z}J_0/r$ 表征的电流，其中 J_0 是常数，r 是距圆柱轴线的径向距离。获得磁场 H 的表达式

(a) $0 \le r \le a$
(b) $r > a$

5.23 对于电流密度 $J = \hat{z}J_0e^{-r}$ 重复问题 5.22

*5.24 在某个导电区域，磁场在柱坐标中由下式给出：

$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{4}{r}[1 - (1 + 3r)e^{-3r}].$$

求电流密度J。

5.25 轴线与 z 轴重合的圆柱形导体，其内部磁场为：

$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{2}{r}[1 - (4r + 1)e^{-4r}] \quad \text{(A/m)} \quad \text{for } r \le a,$$

其中 a 是导体的半径。如果 = 5 cm，导体中流动的总电流是多少?

第 5-4 节：矢量磁势

5.26 参考图5-10：

*(a) 导出 x-y 平面中距导线距离 r 的点 P 处的矢量磁势 A 的表达式。 (b) 从 A 导出 B。证明你的结果与式 (1) 给出的表达式相同。 （5.29），这是通过应用毕奥-萨伐尔定律推导出来的。

5.27 In a given region of space, the vector magnetic potential is given by $\mathbf{A} = \hat{\mathbf{x}}5\cos\pi y + \hat{\mathbf{z}}(2 + \sin\pi x)$ (Wb/m).

*(a) Determine **B**.

(b) Use Eq. (5.66) to calculate the magnetic flux passing through a square loop with 0.25-m long edges if the loop is in the x–y plane, its center is at the origin, and its edges are parallel to the x and y axes.

(c) Calculate Φ again using Eq. (5.67).

5.28 A uniform current density given by

$$\mathbf{J} = \hat{\mathbf{z}}J_0 \qquad (\text{A/m}^2)$$

gives rise to a vector magnetic potential

$$\mathbf{A} = -\hat{\mathbf{z}}\frac{\mu_0 J_0}{4}(x^2 + y^2) \qquad (\text{Wb/m}).$$

(a) Apply the vector Poisson's equation to confirm the above statement.

(b) Use the expression for **A** to find **H**.

(c) Use the expression for **J** in conjunction with Ampère's law to find **H**. Compare your result with that obtained in part (b).

*5.29 A thin current element extending between $z = -L/2$ and $z = L/2$ carries a current I along $+\hat{\mathbf{z}}$ through a circular cross-section of radius a.

(a) Find **A** at a point P located very far from the origin (assume R is so much larger than L that point P may be considered to be at approximately the same distance from every point along the current element).

(b) Determine the corresponding **H**.

Section 5-5: Magnetic Properties of Materials

5.30 In the model of the hydrogen atom proposed by Bohr in 1913, the electron moves around the nucleus at a speed of 2×10^6 m/s in a circular orbit of radius 5×10^{-11} m. What is the magnitude of the magnetic moment generated by the electron's motion?

*5.31 Iron contains 8.5×10^{28} atoms/m^3. At saturation, the alignment of the electrons' spin magnetic moments in iron can contribute 1.5 T to the total magnetic flux density **B**. If the spin magnetic moment of a single electron is 9.27×10^{-24} (A·m^2), how many electrons per atom contribute to the saturated field?

Section 5-6: Magnetic Boundary Conditions

5.32 The x–y plane separates two magnetic media with magnetic permeabilities μ_1 and μ_2 (**Fig. P5.32**). If there is no surface current at the interface and the magnetic field in medium 1 is

$$\mathbf{H}_1 = \hat{\mathbf{x}}H_{1x} + \hat{\mathbf{y}}H_{1y} + \hat{\mathbf{z}}H_{1z},$$

find:

(a) $\mathbf{H}_2$

(b) θ_1 and θ_2

(c) Evaluate $\mathbf{H}_2$, θ_1, and θ_2 for $H_{1x} = 2$ (A/m), $H_{1y} = 0$, $H_{1z} = 4$u (A/m), $\mu_1 = \mu_0$, and $\mu_2 = 4\mu_0$

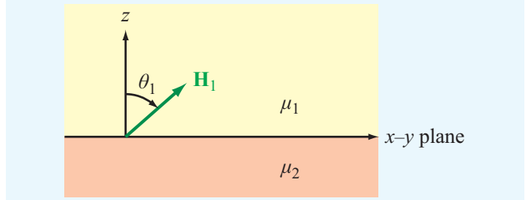


Figure P5.32 Adjacent magnetic media (Problem 5.32).

*5.33 Given that a current sheet with surface current density $\mathbf{J}_s = \hat{\mathbf{x}}8$ (A/m) exists at $y = 0$, the interface between two magnetic media, and $\mathbf{H}_1 = \hat{\mathbf{z}}11$ (A/m) in medium 1 ($y > 0$), determine $\mathbf{H}_2$ in medium 2 ($y < 0$).

5.34 In **Fig. P5.34**, the plane defined by $x - y = 1$ separates medium 1 of permeability μ_1 from medium 2 of permeability μ_2. If no surface current exists on the boundary and

$$\mathbf{B}_1 = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 \qquad (\text{T}),$$

find $\mathbf{B}_2$ and then evaluate your result for $\mu_1 = 5\mu_2$. *Hint:* Start by deriving the equation for the unit vector normal to the given plane.

*5.35 The plane boundary defined by $z = 0$ separates air from a block of iron. If $\mathbf{B}_1 = \hat{\mathbf{x}}4 - \hat{\mathbf{y}}6 + \hat{\mathbf{z}}8$ in air ($z \geq 0$), find $\mathbf{B}_2$ in iron ($z \leq 0$) given that $\mu = 5000\mu_0$ for iron.

5.27 在给定的空间区域中，矢量磁势由 A = x̂5 cosπ y + ẑ (2 + sinπ x) (Wb/m) 给出。

*(a) 确定 B.

(b) 使用等式。(5.66) 计算通过具有 0.25 米长边的方形环的磁通量，如果该环位于 x-y 平面内，其中心位于原点，且其边缘平行于 x 轴和 y 轴。

(c) 使用等式再次计算Φ。（5.67）。

5.28 均匀电流密度由下式给出

$$\mathbf{J} = \hat{\mathbf{z}}J_0 \qquad (\text{A/m}^2)$$

产生矢量磁势

$$\mathbf{A} = -\hat{\mathbf{z}}\frac{\mu_0 J_0}{4}(x^2 + y^2) \qquad (\text{Wb/m}).$$

(a) 应用矢量泊松方程来证实上述说法。(b) 使用 A 的表达式求 H。(c) 使用 J 的表达式结合 Ampère 定律求 H。将你的结果与 (b) 部分得到的结果进行比较。

*5.29 在 z = −L/2 和 z = L/2 之间延伸的细电流元件沿着 +ẑ 通过半径为 a 的圆形横截面承载电流 I。

(a) 在距离原点很远的点 P 处找到 A（假设 R 远大于 L，以至于点 P 可以被认为与沿电流微元的每个点的距离大致相同）。

(b) 确定相应的H.

第 5-5 节：材料的磁性

5.30 在玻尔 1913 年提出的氢原子模型中，电子在半径为 5×10^{-11} m 的圆形轨道上以 2×10^6 m/s 的速度绕原子核运动。电子运动产生的磁矩的大小是多少?

*5.31 铁含有 8.5×10^{28} 原子/m^3。饱和时，铁中电子自旋磁矩的排列可以为总磁通密度 B 贡献 1.5 T。如果单个电子的自旋磁矩为 9.27×10^{-24} (A·m^2)，则每个原子有多少电子对饱和场有贡献?

第 5-6 节：磁边界条件

5.32 x-y 平面将两种磁导率分别为 μ1 和 μ2（的磁介质分开（图 5.32）。P5.32)。如果界面处没有表面电流且介质 1 中的磁场为

$$\mathbf{H}_1 = \hat{\mathbf{x}}H_{1x} + \hat{\mathbf{y}}H_{1y} + \hat{\mathbf{z}}H_{1z},$$

发现：

(a) H_2

(b) θ1 和 θ2

(c) 计算 H_2、θ1 和 θ2，以获得 $H_{1x} = 2$ (A/m)、$H_{1y} = 0$、$H_{1z} = 4$u (A/m)、$\mu_1 = \mu_0$ 和 $\mu_2 = 4\mu_0$

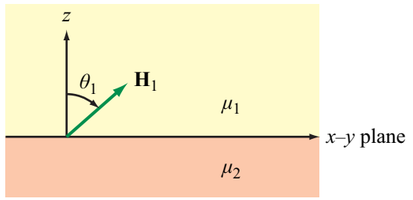


图 P5.32 相邻的磁介质（问题 5.32）。

*5.33 假设在两种磁性介质之间的界面 y = 0 处存在表面电流密度为 Js = x̂ 8 (A/m) 的电流片，以及介质 1 (y > 0) 中的 H1 = ẑ 11 (A/m)，则确定介质 2 (y < 0) 中的 H2。

5.34 在图 P5.34 中，由 x − y = 1 定义的平面将渗透率 μ1 的介质 1 与渗透率 μ2 的介质 2 分开。如果边界上不存在表面电流且

$$\mathbf{B}_1 = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 \qquad (\text{T}),$$

找到 B2，然后评估 μ1 = 5μ2 的结果。提示：首先推导垂直于给定平面的单位向量的方程。

*5.35 由 z = 0 定义的平面边界将空气与铁块分开。如果 B1 = x̂4 − ŷ6 + ẑ8 在空气中 (z ≥ 0)，则在铁 (z ≤ 0) 中找到 B2，前提是铁的 μ = 5000μ0。